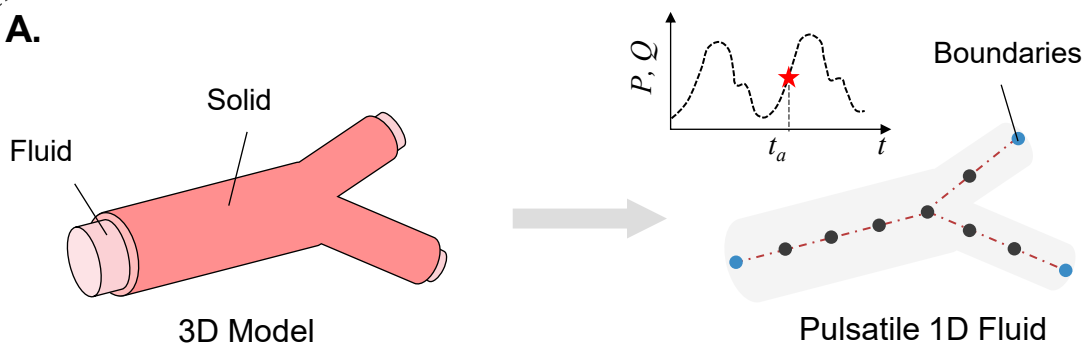
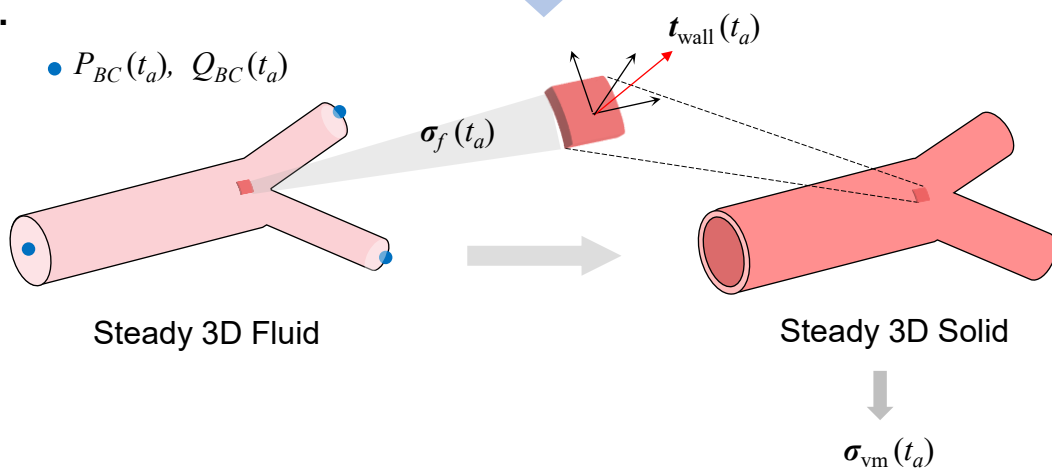
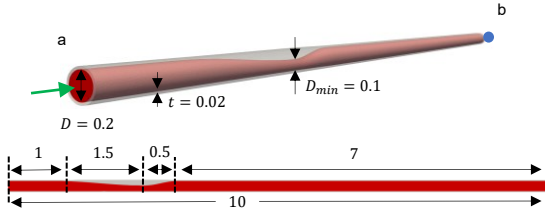
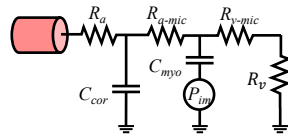
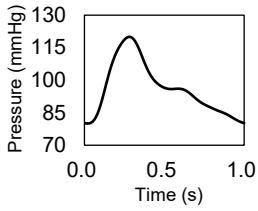


A.**B.**

(A) Ideal coronary artery



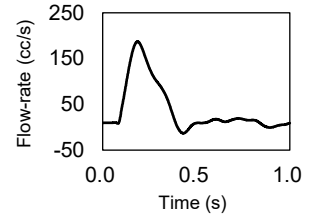
(a). Pressure inlet waveform (b). Coronary boundary model



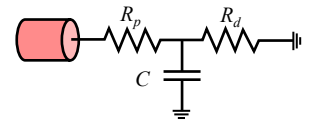
(B) Subject specific AAA



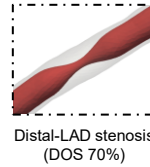
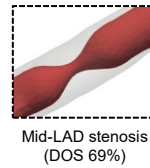
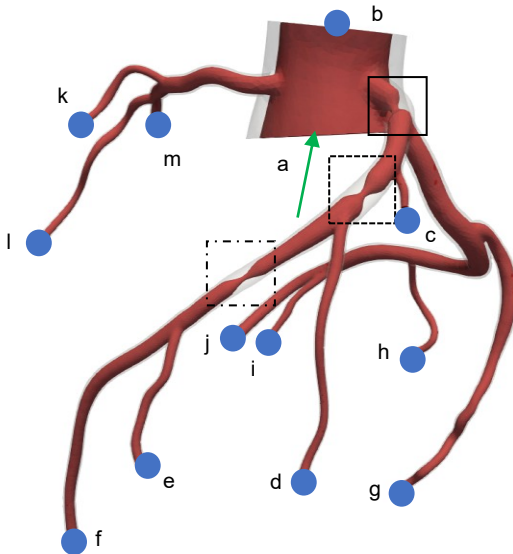
(a). Flow-rate inlet waveform



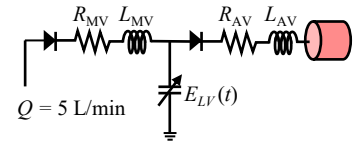
(b,c). Windkessel RCR



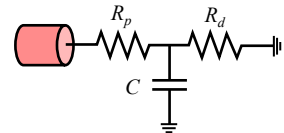
(C) Subject specific coronary artery



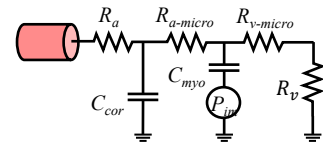
(a). Heart boundary model



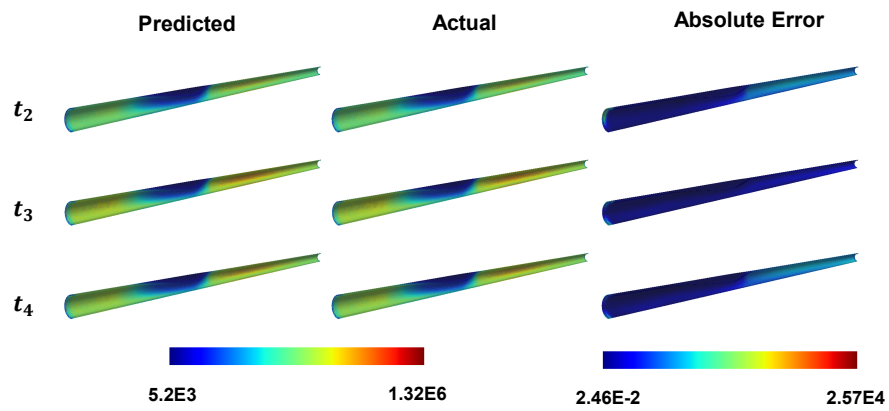
(b). Windkessel RCR



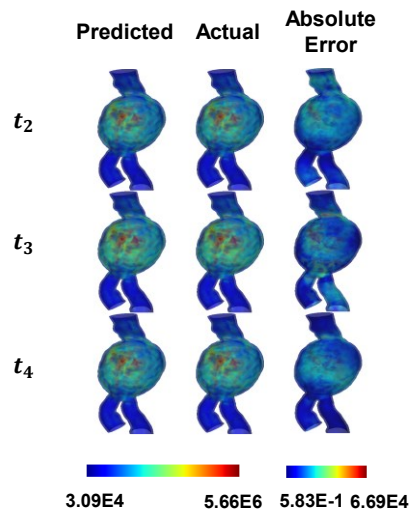
(c~l). Coronary boundary model



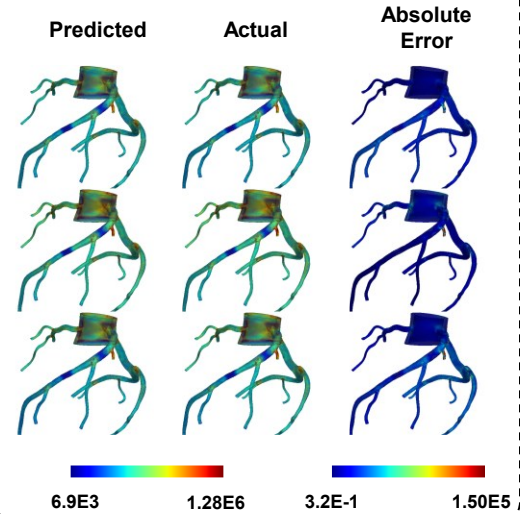
(A) Ideal coronary artery



(B) Subject specific AAA



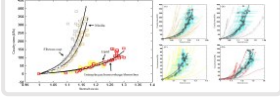
(C) Subject specific coronary artery



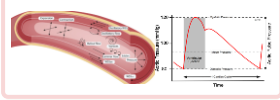
Step 1. Create data set

Input Parameters

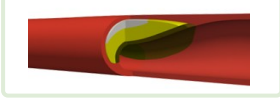
Material properties



Hemodynamic

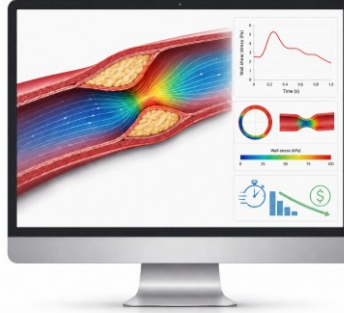


Morphology

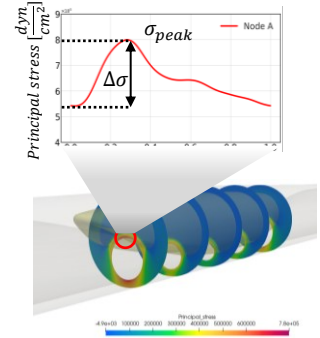


Computational Analysis

Computationally effective FSI simulation



Output Parameters



$$PSS \equiv \max_{i \in N} \sigma_{peak,i}, \Delta PSS \equiv \max_{i \in N} \Delta \sigma_i$$

Step 2. Rupture Vulnerability Index (VI) selection

$$VI \equiv \text{stress}/\text{strength} \\ \text{stress} \in \{PSS, \Delta PSS\}, \quad \text{strength} \propto E_{fc}^\alpha, \quad \alpha \in \{0.0, 0.5, 1.0\}$$

☐ $VI_{clinical}$: VI satisfies all 7 clinical high rupture risk plaque features

$$VI_1 = PSS/E_{fc}^{0.0}$$

$$VI_2 = PSS/E_{fc}^{0.5}$$

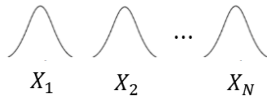
$$VI_3 = PSS/E_{fc}^{1.0}$$

$$VI_4 = \Delta PSS/E_{fc}^{0.0}$$

$$VI_5 = \Delta PSS/E_{fc}^{0.5}$$

$$VI_6 = \Delta PSS/E_{fc}^{1.0}$$

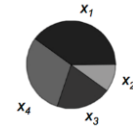
Step 3. Surrogate Model Training & Sensitivity Analysis



Input Parameters

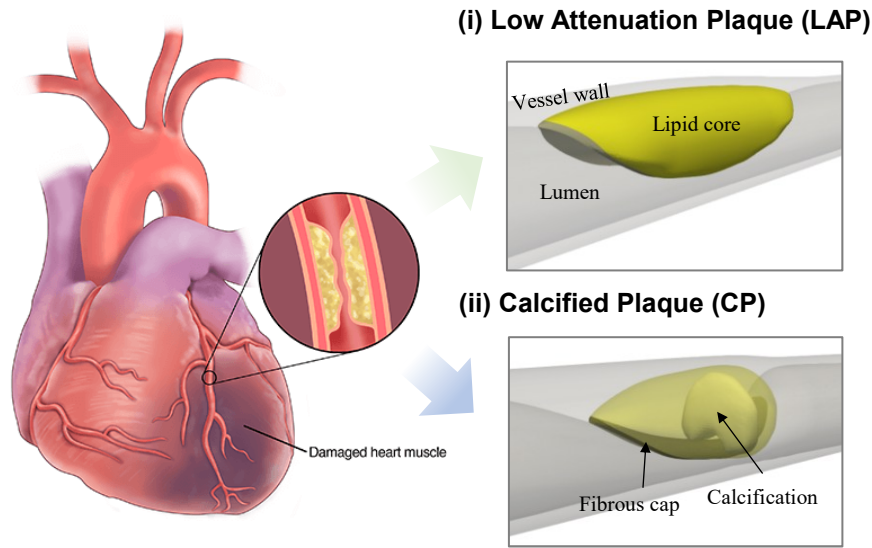
$$VI_{clinical} = f(X_1, \dots, X_N)$$

Surrogate Model

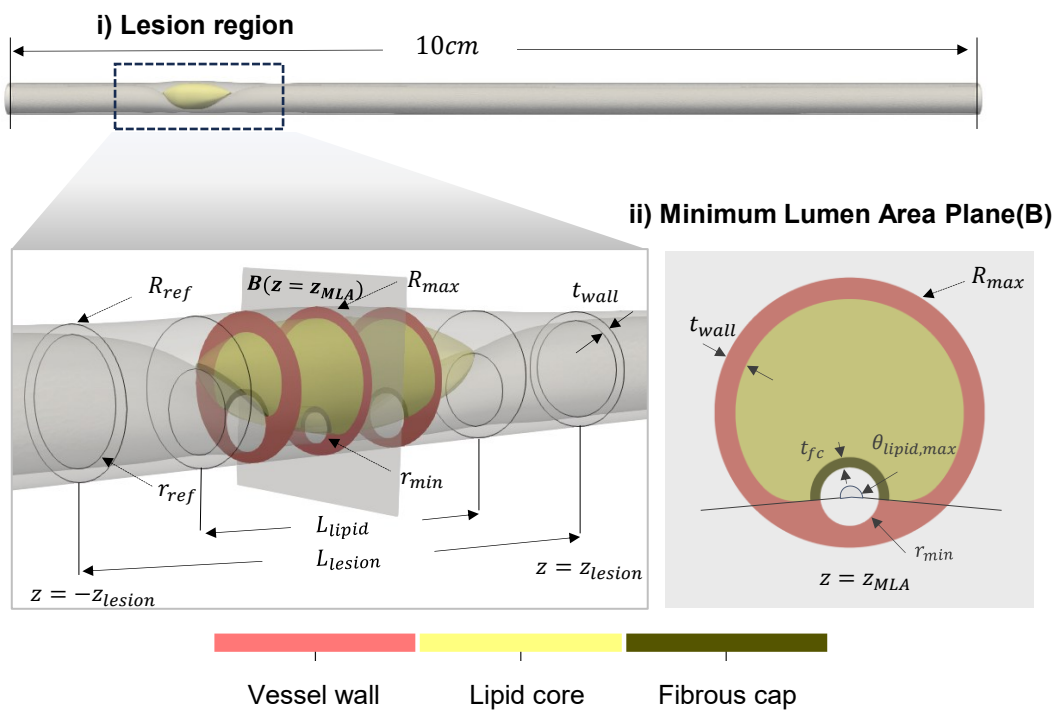


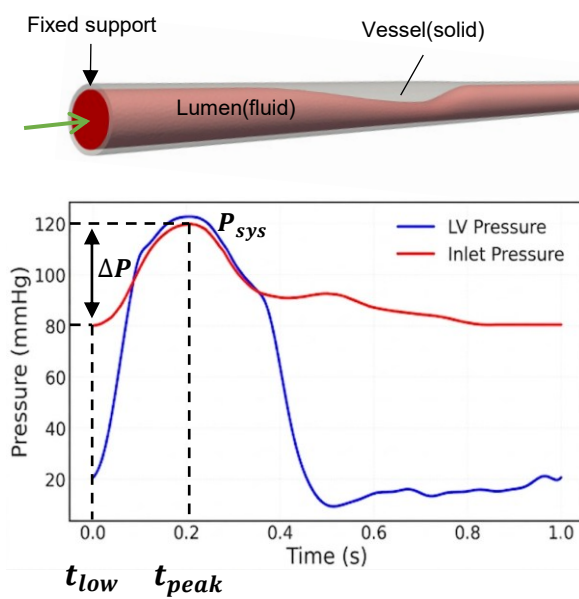
Sensitivity Analysis

A) Plaque classification

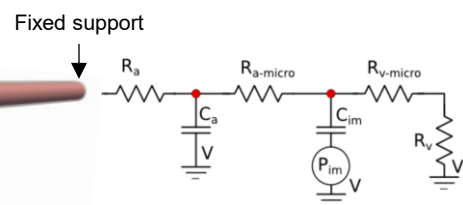


B) Plaque geometry (LAP)





(a) Inlet and Left Ventricular pressure

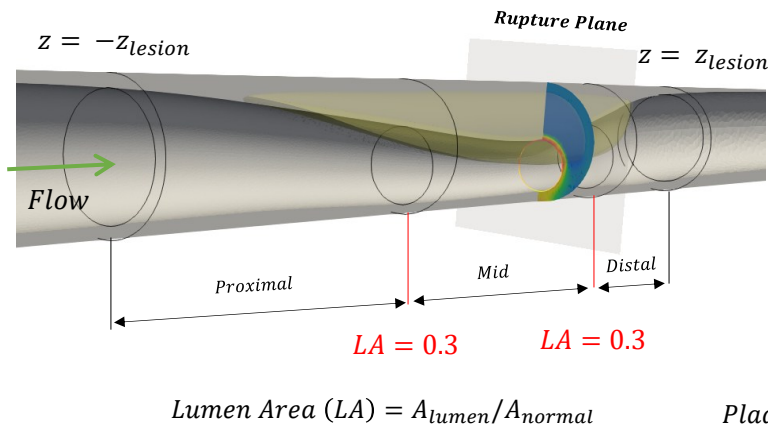


unit: $R[\text{dyn s} / \text{cm}^5]$, $C[\text{cm}^5 / \text{dyn}]$

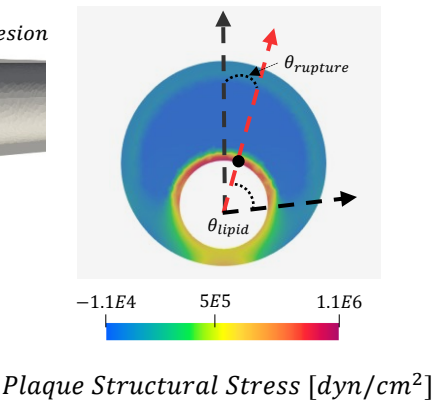
R_a	78,827	C_{im}	3.2E-6
C_a	3.9E-7	R_{v-m}	19,706
R_{a-m}	128,094	R_v	19,706

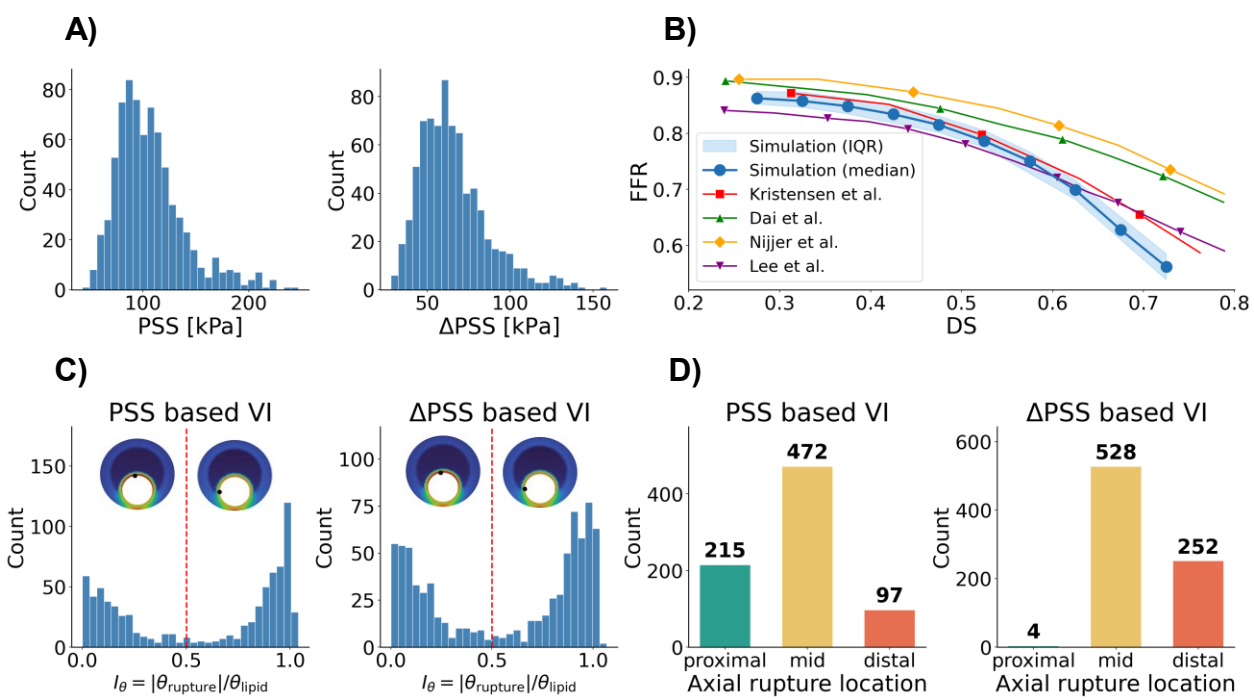
(b) Outlet Coronary Boundary condition

A) Axial Rupture Location

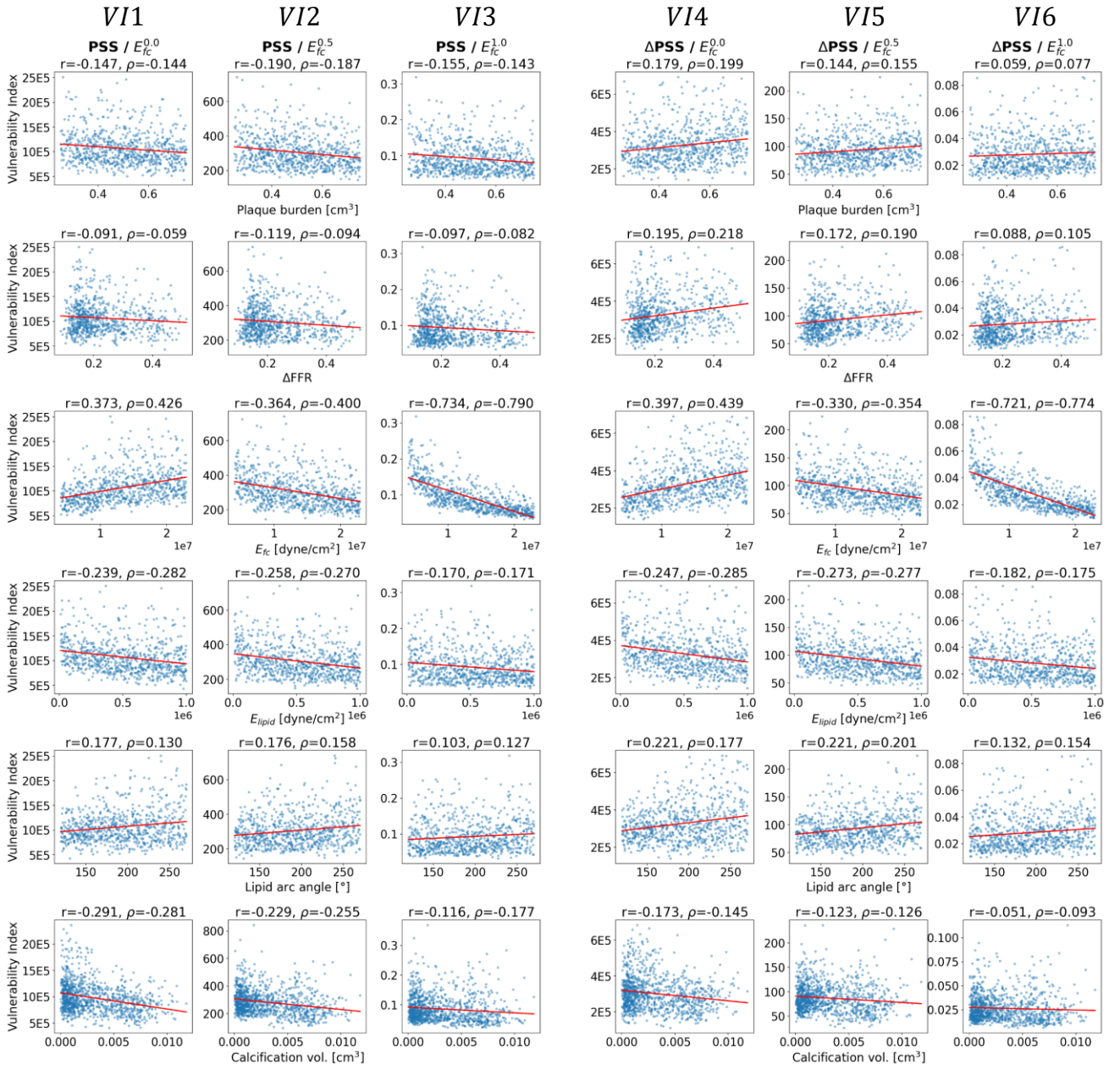


B) Rupture Plane

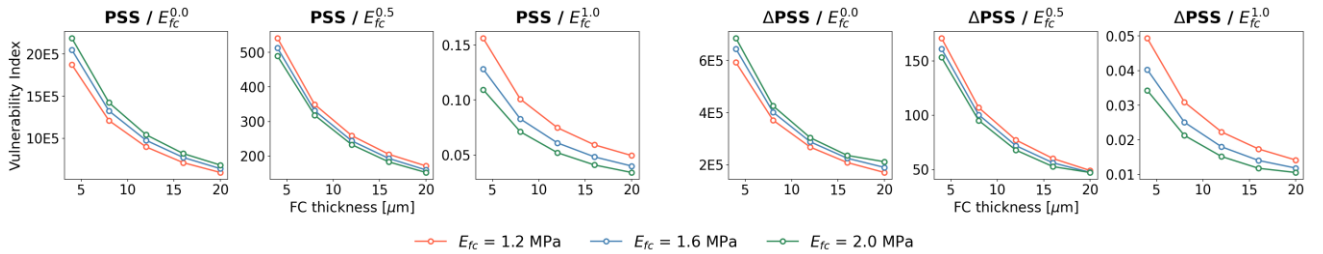




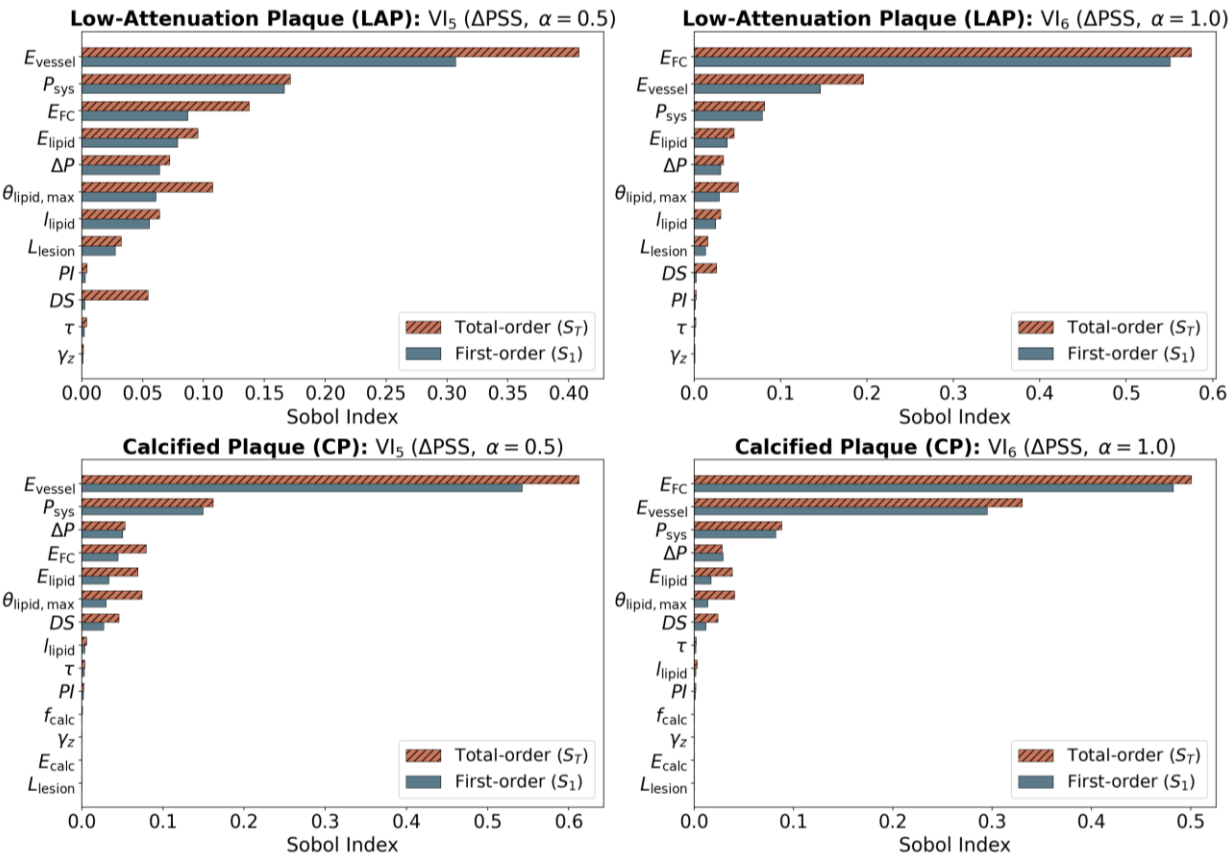
A) LAP and CP data input and Vulnerability Index correlations



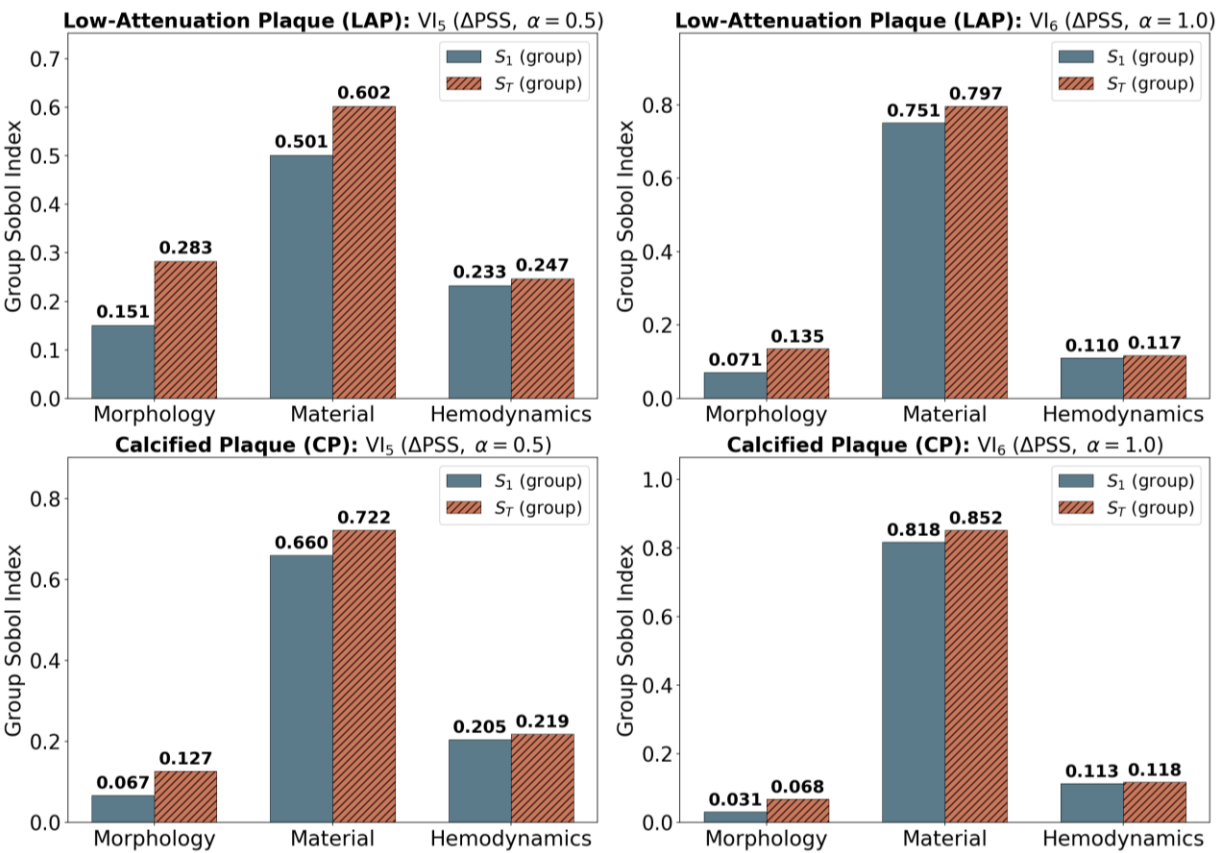
B) Fibrous cap thickness and Vulnerability Index correlations



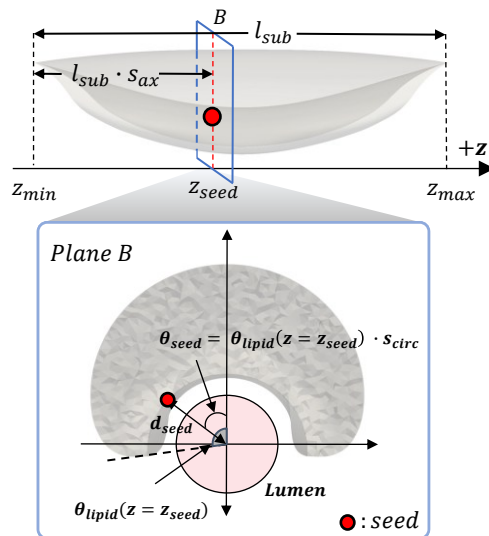
A) Individual Sobol Indices



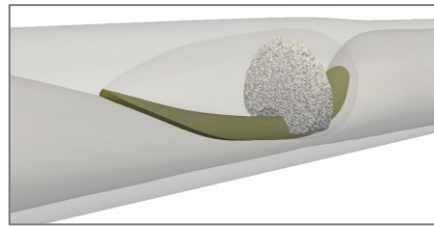
B) Group Sobol Indices



A) Seed location



B) Seed propagation



C) Surface smoothing

